

ELEC4631 Tutorial 6

1. Compute the square root of the following positive definite matrices:

(a) $Q = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$

(b) $Q = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}.$

2. Consider the one state system

$$\dot{x}(t) = x(t) + u(t),$$

and the finite horizon performance index

$$J_{t_f}(x_0, u) = \frac{1}{2} \int_0^{t_f} (x(t)^2 + ru(t)^2) dt.$$

- (a) Write down the RDE that needs to be solved, the optimal control signal, and the optimal finite horizon performance index.
- (b) Let $t_f \rightarrow \infty$ and consider the infinite horizon LQ problem of minimizing the cost

$$J_\infty(x_0, u) = \frac{1}{2} \int_0^\infty (x(t)^2 + ru(t)^2) dt.$$

Solve the CARE, write down the optimal control signal, and the optimal performance index.

3. Determine whether $(Q^{1/2}, A)$ is an observable pair, or not, for the following pairs of Q and A .

(a) $Q = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}.$

(b) $Q = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$.

4. Consider the second order state-space system

$$\dot{x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u.$$

- (a) Suppose that you could access the state. Let $P(t) = \begin{bmatrix} p_1(t) & p_2(t) \\ p_2(t) & p_3(t) \end{bmatrix}$. You want to minimise the finite horizon cost

$$J_{t_f}(x_0, u) = \int_0^{t_f} (x_1(t)^2 + x_2(t)^2 + u(t)^2) dt + x_2(t_f)^2.$$

Write down the equations that must be satisfied by $p_1(t)$, $p_2(t)$ and $p_3(t)$ (including the terminal conditions), write down the optimal control law $u_{\text{opt}}(t)$, and the optimal performance index $J_{t_f, \text{opt}}(x_0)$.

- (b) Suppose that you could access the state. Consider the infinite horizon LQR problem of minimising

$$J_{\infty}(x_0) = \int_0^{\infty} (x_1(t)^2 + x_2(t)^2 + u(t)^2) dt.$$

Is the infinite-horizon LQR problem solvable? If so, write down the CARE for P and compute its positive definite solution, find the optimal feedback gain matrix K , write down the optimal control law $u_{\text{opt}}(t)$, and write down the optimal performance index $J_{\infty, \text{opt}}(x_0)$.